activation of avoidance manoeuvres directly from the spacecraft itself, reducing reliance on ground control. Additionally, drawing parallels with aeronautical regulations, establishing a space traffic management authority becomes crucial to address the congested traffic in LEO and ensure the safety of the space environment.

In summary, safeguarding business continuity and resilience requires the establishment of adaptive systems, the isolation of critical systems, and robust monitoring of cyber activities. Moreover, addressing radiation exposure, space debris, and congestion in space operations necessitates collaborative efforts, technological advancements, and the establishment of regulatory frameworks.

Responsibilities

Space holds strategic importance, and certain satellites are integral to a nation's critical infrastructure. The ramifications of cyber-attacks and threats targeting satellites extend to national security, necessitating commercial actors to comprehend the potential consequences their satellites pose in this regard. Article VI of the Outer Space Treaty (OST) assigns direct responsibility to states for their national space activities. It requires an "appropriate State" to oversee and authorise space activities of non-governmental entities, albeit without precise definitions. The Liability Convention covers physical damage caused by space objects. If harmful interference is attributed to a government or non-government entity, an important question arises: Has the involved state taken sufficient measures to prevent or halt this interference? Failure to take necessary action renders the state responsible for the harmful interference.

However, the notion of responsibility for the use of force can be re-evaluated if a cyber operation reaches a scale and impact comparable to non-cyber operations that constitute a use of force. Establishing a threshold may prove challenging, leading to the consideration of various factors such as severity, immediacy, directness, invasiveness, measurability of effects, military character, state involvement, and presumptive legality. A case-by-case evaluation is crucial in determining whether a cyber operation qualifies for a "use of force." If it meets these parameters, the threat of such an operation would be illegal under international law.

In the event of satellite signal interference or jamming, member states must adhere to International Telecommunication Union (ITU) provisions and engage in cooperation with other states to eliminate harmful interference through bilateral negotiations. If negotiations fail to yield a resolution, the affected state can pursue arbitration. An example illustrating the consequences of a hostile cyber operation is the 1998 US-German ROSAT X-Ray satellite hack. Hackers infiltrated the ground control system at the Goddard Space Flight Centre in Maryland, redirecting the satellite's solar panels towards the sun, resulting in the destruction of batteries and the satellite's subsequent descent to Earth in 2011.

With the framework governing state responsibility in place, risk mitigation in space operations relies on defining and understanding the vulnerabilities of the most attractive assets to hackers, both physical and virtual, and implementing adequate defence and protection measures. The National Institute of Standards and Technology (NIST) Cybersecurity Framework (CSF) facilitates the establishment of a risk management framework (RMF), already adopted by several governments for cybersecurity. The NIST CSF encompasses five elements: identification, protection, detection, response, and recovery, which are implemented simultaneously to form the foundation for robust cybersecurity risk management. Additionally, the United States Department of Defence (DoD) introduced the Cybersecurity Maturity Model Certification (CMMC) to evaluate the cybersecurity capabilities, readiness, and sophistication of defence contractors. This serves as an exemplary model for effective cybersecurity practices. Overall, these frameworks and initiatives showcase successful approaches to managing cybersecurity risks and safeguarding assets in space.

Final thoughts

The emergence of extensive networks of satellites in Low Earth Orbit (LEO) presents a range of intricate technical and legal obstacles, with cybersecurity standing out as a paramount concern. These vast constellations are reshaping the landscape of satellite communications, offering ease of deployment and utilisation. This study proposes potential technologies to mitigate cybersecurity risks and ensure the security of space assets. Moreover, to safeguard space assets and the critical infrastructure supporting them against cyber threats, while guaranteeing the uninterrupted flow of space operations for both governmental and commercial entities, this work re-examines the legal definition of large constellations and the associated responsibilities in risk mitigation. Finally, this exploration presents successful instances of national frameworks or organisations as a valuable reference for future endeavors to establish a legal framework tailored to space activities involving expansive constellations. EFY